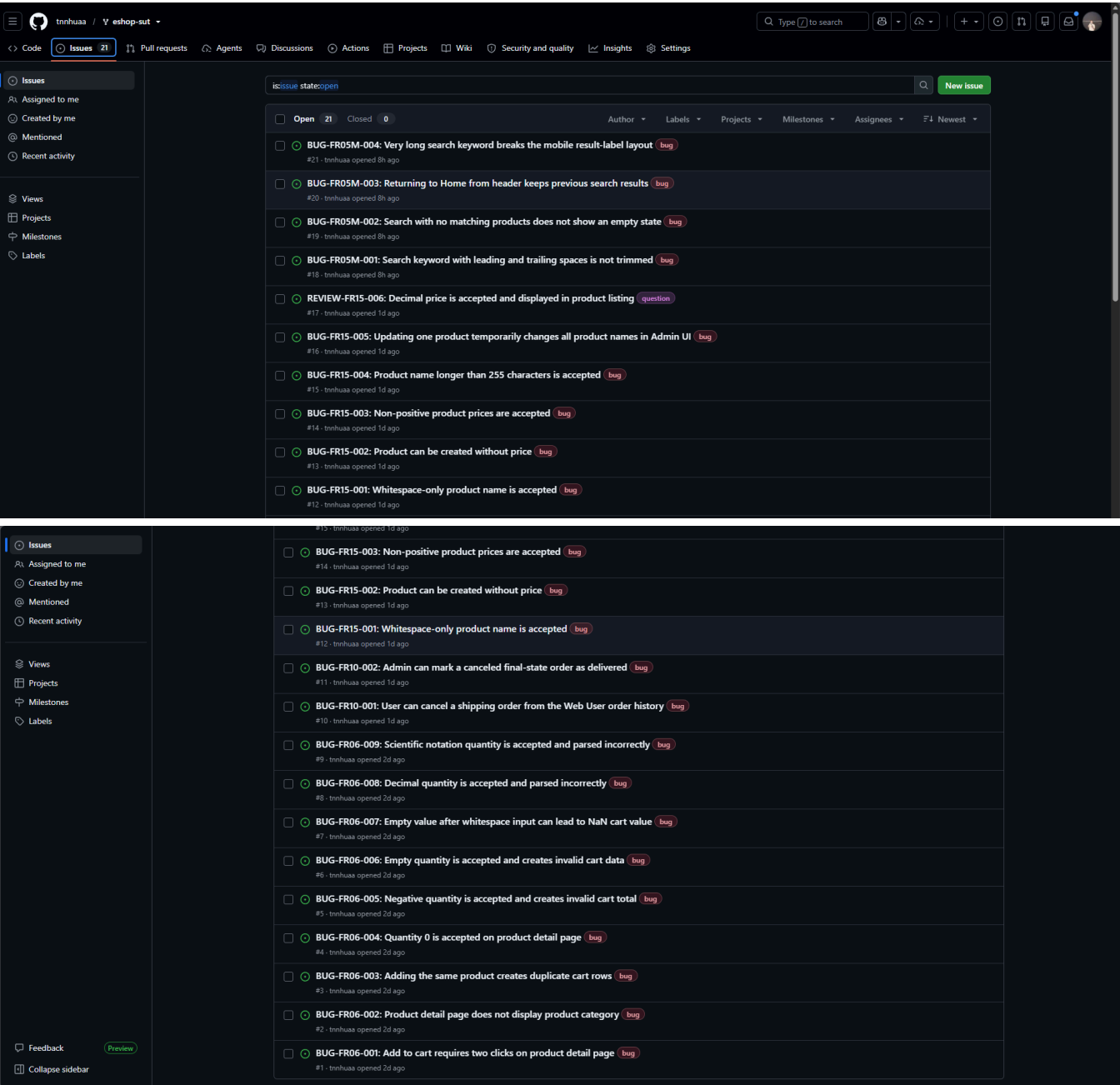


HW02 Bug Report

Bugs on the GitHub Issues page



Bug Summary

Bug ID	Feature	Title	Severity	Status	GitHub Issue	Evidence
BUG-FR06-001	FR-06 Product Detail View	Add to cart requires two clicks on product detail page	Major	Open	#1	features/FR06_Product_Detail_Web/evidence/BUG-FR06-001.png ; BUG-FR06-001_2.png

Bug ID	Feature	Title	Severity	Status	GitHub Issue	Evidence
BUG-FR06-002	FR-06 Product Detail View	Product detail page does not display product category	Medium	Open	#2	features/FR06_Product_Detail_Web/evidence/BUG-FR06-002.png
BUG-FR06-003	FR-06 Product Detail View / Cart Integration	Adding the same product creates duplicate cart rows	Medium	Open	#3	features/FR06_Product_Detail_Web/evidence/BUG-FR06-003.png
BUG-FR06-004	FR-06 Product Detail View	Quantity 0 is accepted on product detail page	Major	Open	#4	features/FR06_Product_Detail_Web/evidence/BUG-FR06-004.png
BUG-FR06-005	FR-06 Product Detail View	Negative quantity is accepted and creates invalid cart total	Major	Open	#5	features/FR06_Product_Detail_Web/evidence/BUG-FR06-005.png
BUG-FR06-006	FR-06 Product Detail View	Empty quantity is accepted and creates invalid cart data	Major	Open	#6	features/FR06_Product_Detail_Web/evidence/BUG-FR06-006.png
BUG-FR06-007	FR-06 Product Detail View	Empty value after whitespace input can lead to NaN cart value	Major	Open	#7	features/FR06_Product_Detail_Web/evidence/BUG-FR06-007.png
BUG-FR06-008	FR-06 Product Detail View	Decimal quantity is accepted and parsed incorrectly	Major	Open	#8	features/FR06_Product_Detail_Web/evidence/BUG-FR06-008.png
BUG-FR06-009	FR-06 Product Detail View	Scientific notation quantity is accepted and parsed incorrectly	Major	Open	#9	features/FR06_Product_Detail_Web/evidence/BUG-FR06-009.png ; BUG-FR06-009_2.png

Bug ID	Feature	Title	Severity	Status	GitHub Issue	Evidence
BUG-FR10-001	FR-10 Order State Machine	User can cancel a shipping order from the Web User order history	Major	Open	#10	features/FR10_Order_State_Machine/evidence/BUG-FR10-001-01.png ; BUG-FR10-001-02.png
BUG-FR10-002	FR-10 Order State Machine	Admin can mark a canceled final-state order as delivered	Major	Open	#11	features/FR10_Order_State_Machine/evidence/BUG-FR10-002-01.png ; BUG-FR10-002-02.png
BUG-FR15-001	FR-15 Product CRUD Admin	Whitespace-only product name is accepted	Major	Open	#12	features/FR15_Product_CRUD_Admin/evidence/BUG-FR15-001.png
BUG-FR15-002	FR-15 Product CRUD Admin	Product can be created without price	Major	Open	#13	features/FR15_Product_CRUD_Admin/evidence/BUG-FR15-002.png
BUG-FR15-003	FR-15 Product CRUD Admin	Non-positive product prices are accepted	Major	Open	#14	features/FR15_Product_CRUD_Admin/evidence/BUG-FR15-003.png
BUG-FR15-004	FR-15 Product CRUD Admin	Product name longer than 255 characters is accepted	Major	Open	#15	features/FR15_Product_CRUD_Admin/evidence/BUG-FR15-004.png
BUG-FR15-005	FR-15 Product CRUD Admin	Updating one product temporarily changes all product names in Admin UI	Major	Open	#16	features/FR15_Product_CRUD_Admin/evidence/BUG-FR15-005-1.png ; features/FR15_Product_CRUD_Admin/evidence/BUG-FR15-005-2.png ; features/FR15_Product_CRUD_Admin/evidence/BUG-FR15-005-3.png
REVIEW-FR15-006	FR-15 Product CRUD Admin	Decimal price is accepted and displayed in product listing	Medium	Open / Needs confirmation	#17	features/FR15_Product_CRUD_Admin/evidence/BUG-FR15-006.png

Bug ID	Feature	Title	Severity	Status	GitHub Issue	Evidence
BUG-FR05M-001	FR-05-M Product Listing and Search Mobile	Search keyword with leading and trailing spaces is not trimmed	Medium	Open	#18	features/FR05_Product_Search_Mobile/evidence/BUG-FR05M-001-1.jpg ; features/FR05_Product_Search_Mobile/evidence/BUG-FR05M-001-2.jpg
BUG-FR05M-002	FR-05-M Product Listing and Search Mobile	Search with no matching products does not show an empty state	Major	Open	#19	features/FR05_Product_Search_Mobile/evidence/BUG-FR05M-002.jpg
BUG-FR05M-003	FR-05-M Product Listing and Search Mobile	Returning to Home from header keeps previous search results	Medium	Open	#20	GitHub Issue #20
BUG-FR05M-004	FR-05-M Product Listing and Search Mobile	Very long search keyword breaks the mobile result-label layout	Medium	Open	#21	features/FR05_Product_Search_Mobile/evidence/BUG-FR05M-004.jpg

Environment

- App: Web User
- Main page under test: Product Detail
- Main route: /product/1
- SUT commit hash: 85af3ba875c88283615e22cb108f13e2fccaf0e9
- Execution type: Manual UI testing
- Evidence storage: features/FR06_Product_Detail_Web/evidence/

Detailed Bug Reports

BUG-FR06-001: Add to cart requires two clicks on product detail page

- Feature: FR-06 Product Detail View
- Related test cases: FR06-DT-006, FR06-DT-015
- Severity: Major
- Priority: High
- GitHub issue: #1
- Preconditions: Web User app and backend are running; product ID 1 exists.
- Steps to reproduce:
 1. Open /product/1 .

2. Enter quantity **1**.
 3. Click **Add to Cart** once.
 4. Open the cart page.
- Expected result: The product is added to the cart after one valid click and the user receives visible feedback.
 - Actual result: The first click does not add the product. The product is added only after clicking **Add to Cart** a second time.
 - Evidence:
 - `features/FR06_Product_Detail_Web/evidence/BUG-FR06-001.png`
 - `features/FR06_Product_Detail_Web/evidence/BUG-FR06-001_2.png`
 - Notes: This issue affects other add-to-cart tests, so later quantity tests used the known second-click workaround to isolate quantity validation behavior.

BUG-FR06-002: Product detail page does not display product category

- Feature: FR-06 Product Detail View
- Related test case: FR06-DT-001
- Severity: Medium
- Priority: Medium
- GitHub issue: **#2**
- Preconditions: Web User app and backend are running; product ID **1** exists.
- Steps to reproduce:
 1. Open `/product/1`.
 2. Observe product information on the detail page.
- Expected result: Product image, name, price, description, and category are visible.
- Actual result: The page shows image, name, price, and description, but no visible category information is shown.
- Evidence: `features/FR06_Product_Detail_Web/evidence/BUG-FR06-002.png`
- Notes: FR-06 requires category display. The requirement does not specify whether category name or ID is required, but no category information is visible.

BUG-FR06-003: Adding the same product creates duplicate cart rows

- Feature: FR-06 Product Detail View / FR-07 Cart Integration
- Related test case: FR06-DT-008
- Severity: Medium
- Priority: Medium
- GitHub issue: **#3**
- Preconditions: Web User app and backend are running; product ID **1** exists.
- Steps to reproduce:
 1. Open `/product/1`.
 2. Enter quantity **1**.
 3. Click **Add to Cart** until the product is added.
 4. Return to `/product/1`.
 5. Enter quantity **2**.
 6. Click **Add to Cart** until the product is added.
 7. Open the cart page.
- Expected result: The existing cart row for the same product increases from quantity **1** to **3**.
- Actual result: The cart shows duplicate rows for the same product instead of merging quantity.
- Evidence: `features/FR06_Product_Detail_Web/evidence/BUG-FR06-003.png`
- Notes: This is triggered from FR-06 product detail but is related to cart behavior defined by FR-07.

BUG-FR06-004: Quantity 0 is accepted on product detail page

- Feature: FR-06 Product Detail View
- Related test case: FR06-BVA-001
- Severity: Major
- Priority: High
- GitHub issue: #4
- Preconditions: Web User app and backend are running; product ID 1 exists.
- Steps to reproduce:
 1. Open /product/1 .
 2. Enter quantity 0 .
 3. Click **Add to Cart** until the product is added.
 4. Open the cart page.
- Expected result: Quantity below minimum 1 is rejected and the cart remains unchanged.
- Actual result: Quantity 0 is accepted and added to the cart.
- Evidence: features/FR06_Product_Detail_Web/evidence/BUG-FR06-004.png
- Notes: This violates the explicit FR-06 minimum quantity boundary.

BUG-FR06-005: Negative quantity is accepted and creates invalid cart total

- Feature: FR-06 Product Detail View
- Related test case: FR06-BVA-004
- Severity: Major
- Priority: High
- GitHub issue: #5
- Preconditions: Web User app and backend are running; product ID 1 exists.
- Steps to reproduce:
 1. Open /product/1 .
 2. Enter quantity -1 .
 3. Click **Add to Cart** until the product is added.
 4. Open the cart page.
- Expected result: Negative quantity is rejected and the cart remains unchanged.
- Actual result: Negative quantity is accepted and can create negative cart quantity/total.
- Evidence: features/FR06_Product_Detail_Web/evidence/BUG-FR06-005.png
- Notes: This is a high-risk quantity validation defect because it can corrupt cart totals.

BUG-FR06-006: Empty quantity is accepted and creates invalid cart data

- Feature: FR-06 Product Detail View
- Related test case: FR06-DT-010
- Severity: Major
- Priority: High
- GitHub issue: #6
- Preconditions: Web User app and backend are running; product ID 1 exists.
- Steps to reproduce:
 1. Open /product/1 .
 2. Clear the quantity field so it is empty.
 3. Click **Add to Cart** until the product is added.
 4. Open the cart page.
- Expected result: Empty quantity is rejected and the cart remains unchanged.
- Actual result: Empty quantity is accepted and can create invalid cart quantity/total.

- Evidence: [features/FR06_Product_Detail_Web/evidence/BUG-FR06-006.png](#)
- Notes: This is separated from whitespace because it is a direct empty-input class.

BUG-FR06-007: Empty value after whitespace input can lead to NaN cart value

- Feature: FR-06 Product Detail View
- Related test case: FR06-DT-011
- Severity: Major
- Priority: High
- GitHub issue: [#7](#)
- Preconditions: Web User app and backend are running; product ID [1](#) exists.
- Steps to reproduce:
 1. Open [/product/1](#).
 2. Try entering spaces into the quantity input.
 3. Observe that spaces are blocked or the field becomes empty.
 4. Click **Add to Cart** until the product is added.
 5. Open the cart page.
- Expected result: Spaces-only or empty-after-space quantity is rejected and the cart remains unchanged.
- Actual result: The resulting empty/invalid value can still be added and may produce invalid or NaN cart value.
- Evidence: [features/FR06_Product_Detail_Web/evidence/BUG-FR06-007.png](#)
- Notes: This issue is tracked separately from direct empty quantity because the user action begins with whitespace input.

BUG-FR06-008: Decimal quantity is accepted and parsed incorrectly

- Feature: FR-06 Product Detail View
- Related test case: FR06-BVA-005
- Severity: Major
- Priority: High
- GitHub issue: [#8](#)
- Preconditions: Web User app and backend are running; product ID [1](#) exists.
- Steps to reproduce:
 1. Open [/product/1](#).
 2. Enter quantity [1.5](#).
 3. Click **Add to Cart** until the product is added.
 4. Open the cart page.
- Expected result: Decimal quantity is rejected because FR-06 only allows positive integer quantity.
- Actual result: Decimal quantity [1.5](#) is accepted and parsed as quantity [1](#) in the cart.
- Evidence: [features/FR06_Product_Detail_Web/evidence/BUG-FR06-008.png](#)
- Notes: This is a decimal input class and is tracked separately from scientific notation.

BUG-FR06-009: Scientific notation quantity is accepted and parsed incorrectly

- Feature: FR-06 Product Detail View
- Related test case: FR06-DT-012
- Severity: Major
- Priority: High
- GitHub issue: [#9](#)
- Preconditions: Web User app and backend are running; product ID [1](#) exists.
- Steps to reproduce:
 1. Open [/product/1](#).
 2. Enter quantity [1e2](#).
 3. Click **Add to Cart** until the product is added.

4. Open the cart page.

- Expected result: Scientific notation is rejected because FR-06 only allows positive integer quantity.
- Actual result: Scientific notation value `1e2` is accepted by the quantity input and parsed incorrectly as the leading integer value instead of being rejected.
- Evidence:
 - `features/FR06_Product_Detail_Web/evidence/BUG-FR06-009.png`
 - `features/FR06_Product_Detail_Web/evidence/BUG-FR06-009_2.png`
- Notes: This is a scientific-notation input class and is tracked separately from decimal input.

BUG-FR10-001: User can cancel a shipping order from the Web User order history

- Feature: FR-10 Order State Machine
- Related test cases: FR10-DT-006, FR10-DT-021, FR10-BVA-002
- Severity: Major
- Priority: High
- GitHub issue: `#10`
- Preconditions: Web User app and backend are running; a normal user is logged in; at least one order exists with status `shipping` / `Đang giao`.
- Steps to reproduce:
 1. Log in to Web User as a normal user.
 2. Open Profile / Order History.
 3. Find an order whose status is `Đang giao`.
 4. Observe that the **Hủy đơn** button is visible.
 5. Click **Hủy đơn**.
 6. Observe the order status after the action.
- Expected result: User cannot cancel an order in `shipping` state. The UI should not show **Hủy đơn**, and the order should remain `Đang giao`.
- Actual result: Web User UI shows **Hủy đơn** for a shipping order. Clicking it succeeds and changes the order status to `Đã hủy`.
- Evidence:
 - `features/FR10_Order_State_Machine/evidence/BUG-FR10-001-01.png`
 - `features/FR10_Order_State_Machine/evidence/BUG-FR10-001-02.png`
- Notes: This violates the FR-10 boundary between `confirmed`, where user cancellation is still allowed, and `shipping`, where user cancellation should be forbidden.

BUG-FR10-002: Admin can mark a canceled final-state order as delivered

- Feature: FR-10 Order State Machine
- Related test cases: FR10-DT-012, FR10-DT-021, FR10-DT-022, FR10-DT-023, FR10-BVA-004
- Severity: Major
- Priority: High
- GitHub issue: `#11`
- Preconditions: Admin Web app and backend are running; Admin is logged in; at least one order exists with status `canceled` / `Đã hủy`.
- Steps to reproduce:
 1. Log in to Admin Web.
 2. Open Order Management.
 3. Find an order whose status is `Đã hủy`.
 4. Observe that **Đánh dấu Đã giao** is visible.
 5. Click **Đánh dấu Đã giao**.
 6. Reopen Admin Order Management and Web User Order History.

- Expected result: `canceled` is a final state. Admin UI should not expose any outgoing transition from `Đã hủy`, and the order should remain `Đã hủy`.
- Actual result: Admin UI shows **Đánh dấu Đã giao** for a canceled order. Clicking it changes the order to `Đã giao`, and the changed state is visible from both Admin and Web User UI.
- Evidence:
 - `features/FR10_Order_State_Machine/evidence/BUG-FR10-002-01.png`
 - `features/FR10_Order_State_Machine/evidence/BUG-FR10-002-02.png`
 - `features/FR10_Order_State_Machine/evidence/BUG-FR10-002-03.png`
- Notes: This violates final-state protection in the FR-10 state machine.

BUG-FR15-001: Whitespace-only product name is accepted

- Feature: FR-15 Product CRUD Admin
- Related test case: FR15-DT-008
- Severity: Major
- Priority: High
- GitHub issue: `#12`
- Preconditions: Admin Web and backend are running; Admin is logged in.
- Steps to reproduce:
 1. Open Admin Product Management.
 2. Create a product with name containing only spaces.
 3. Enter a valid positive price and select an existing category.
 4. Save the product.
- Expected result: Whitespace-only name is treated as empty after trimming and the product is rejected.
- Actual result: The product is accepted and appears in the Admin product list.
- Evidence: `features/FR15_Product_CRUD_Admin/evidence/BUG-FR15-001.png`
- Notes: FR-15 requires product name to be mandatory. A spaces-only value should not satisfy the required-name rule.

BUG-FR15-002: Product can be created without price

- Feature: FR-15 Product CRUD Admin
- Related test cases: FR15-DT-025, FR15-BVA-007
- Severity: Major
- Priority: High
- GitHub issue: `#13`
- Preconditions: Admin Web and backend are running; Admin is logged in.
- Steps to reproduce:
 1. Open Admin Product Management.
 2. Enter a valid product name.
 3. Leave price empty.
 4. Select an existing category.
 5. Save the product.
- Expected result: Product is rejected because price is required.
- Actual result: Product is created and appears in the product list.
- Evidence: `features/FR15_Product_CRUD_Admin/evidence/BUG-FR15-002.png`
- Notes: Browser/UI validation is not sufficient because empty price still reaches the save behavior.

BUG-FR15-003: Non-positive product prices are accepted

- Feature: FR-15 Product CRUD Admin
- Related test cases: FR15-BVA-008, FR15-BVA-009
- Severity: Major

- Priority: High
- GitHub issue: [#14](#)
- Preconditions: Admin Web and backend are running; Admin is logged in.
- Steps to reproduce:
 1. Open Admin Product Management.
 2. Create a product with valid name and category.
 3. Enter price , save, and observe the list.
 4. Repeat with price .
- Expected result: Prices less than or equal to are rejected because price must be positive ().
- Actual result: Products with negative and zero prices are accepted and displayed.
- Evidence: [features/FR15_Product_CRUD_Admin/evidence/BUG-FR15-003.png](#)
- Notes: This violates the explicit positive-price boundary.

BUG-FR15-004: Product name longer than 255 characters is accepted

- Feature: FR-15 Product CRUD Admin
- Related test cases: FR15-BVA-006, FR15-BVA-015
- Severity: Major
- Priority: High
- GitHub issue: [#15](#)
- Preconditions: Admin Web and backend are running; Admin is logged in.
- Steps to reproduce:
 1. Open Admin Product Management.
 2. Create or edit a product using a 256-character ASCII name.
 3. Enter a valid price and select an existing category.
 4. Save the product.
- Expected result: Product is rejected because the name exceeds the maximum 255-character limit.
- Actual result: The 256-character product name is accepted and displayed.
- Evidence: [features/FR15_Product_CRUD_Admin/evidence/BUG-FR15-004.png](#)
- Notes: This violates the upper boundary for product name length.

BUG-FR15-005: Updating one product temporarily changes all product names in Admin UI

- Feature: FR-15 Product CRUD Admin
- Related test cases: FR15-DT-013, FR15-DT-019, FR15-DT-031, FR15-BVA-014
- Severity: Major
- Priority: High
- GitHub issue: [#16](#)
- Preconditions: Admin Web and backend are running; Admin is logged in; Product A and Product B exist with different names.
- Steps to reproduce:
 1. Open Admin Product Management.
 2. Record Product A and Product B names.
 3. Edit only Product A and save a new name.
 4. Observe the product list immediately after save.
 5. Reload the page and observe the product list again.
- Expected result: Only Product A changes. Product B and other products remain unchanged immediately after save and after reload.
- Actual result: Immediately after save, many rows display Product A's new name and Product B is hidden by the incorrect UI state. After reload, persisted data shows only Product A changed.
- Evidence:
 - [features/FR15_Product_CRUD_Admin/evidence/BUG-FR15-005-1.png](#)
 - [features/FR15_Product_CRUD_Admin/evidence/BUG-FR15-005-2.png](#)

- `features/FR15_Product_CRUD_Admin/evidence/BUG-FR15-005-3.png`
- Notes: This violates the UI-level interpretation of "only the edited product changes", even though the persisted backend state appears correct after reload.

REVIEW-FR15-006: Decimal price is accepted and displayed in product listing

- Feature: FR-15 Product CRUD Admin
- Related test case: FR15-BVA-010
- Severity: Medium if integer-only VND is confirmed; otherwise not a defect
- Priority: Medium
- GitHub issue: `#17`
- Preconditions: Admin Web and backend are running; Admin is logged in.
- Steps to reproduce:
 1. Open Admin Product Management.
 2. Enter a valid product name.
 3. Enter price `0.01`.
 4. Select an existing category.
 5. Save the product.
- Expected result: The system handles product price consistently with the requirement. Since FR-15 only says price must be a positive number greater than `0`, decimal `0.01` may be considered valid. If product price is intended to be integer-only VND, the UI should reject decimal prices clearly.
- Actual result: Admin UI allows creating a product with decimal price `0.01`. After saving, the product appears in the product list and the price is displayed as `0,01 VND`.
- Evidence: `features/FR15_Product_CRUD_Admin/evidence/BUG-FR15-006.png`
- Notes: This item is a requirement clarification/potential defect, not a confirmed defect under the current `> 0` wording.

BUG-FR05M-001: Search keyword with leading and trailing spaces is not trimmed

- Feature: FR-05-M Product Listing and Search Mobile
- Related test case: FR05M-DT-005
- Severity: Medium
- Priority: Medium
- GitHub issue: `#18`
- Preconditions: Backend and React Native Mobile app are running; FR05M seed products exist.
- Steps to reproduce:
 1. Open the mobile app Home screen.
 2. Enter a valid matching keyword with leading/trailing spaces, such as `FR05M-ALPHA`.
 3. Tap Search.
 4. Compare the result with searching the trimmed keyword `FR05M-ALPHA`.
- Expected result: The search keyword is trimmed before searching. `FR05M-ALPHA` returns the same matching products as `FR05M-ALPHA`.
- Actual result: The keyword with leading/trailing spaces is not trimmed, so the app does not return the same result as the trimmed keyword.
- Evidence:
 - `features/FR05_Product_Search_Mobile/evidence/BUG-FR05M-001-1.jpg`
 - `features/FR05_Product_Search_Mobile/evidence/BUG-FR05M-001-2.jpg`
- Notes: Trimming is recorded as a reviewed search-semantics assumption because FR-05 does not explicitly define spaces around keywords.

BUG-FR05M-002: Search with no matching products does not show an empty state

- Feature: FR-05-M Product Listing and Search Mobile

- Related test cases: FR05M-DT-007, FR05M-BVA-003
- Severity: Major
- Priority: High
- GitHub issue: [#19](#)
- Preconditions: Backend and React Native Mobile app are running.
- Steps to reproduce:
 1. Open the mobile app Home screen.
 2. Enter a keyword that returns no products, such as [FR05M-NO-MATCH-XYZ](#).
 3. Tap Search.
 4. Observe the result area.
- Expected result: When no products match the search keyword, the app shows a suitable empty-state/no-result message.
- Actual result: The mobile app shows an empty/blank result area without a clear no-result message.
- Evidence: [features/FR05_Product_Search_Mobile/evidence/BUG-FR05M-002.jpg](#)
- Notes: This is a confirmed FR-05-M requirement defect because FR-05 explicitly requires an empty state when there is no search result.

BUG-FR05M-003: Returning to Home from header keeps previous search results

- Feature: FR-05-M Product Listing and Search Mobile
- Related test case: FR05M-DT-018
- Severity: Medium
- Priority: Medium
- GitHub issue: [#20](#)
- Preconditions: Backend and React Native Mobile app are running; search result is currently filtered.
- Steps to reproduce:
 1. Open the mobile app Home screen.
 2. Search for a keyword that filters the product list.
 3. Tap the app header/logo to return to Home.
 4. Observe the product list shown on the Home screen.
- Expected result: Returning to Home from the header/logo resets the search state and shows the default/all product listing, equivalent to an empty search.
- Actual result: The Home screen still shows the previous search results instead of resetting to all products.
- Evidence: GitHub Issue [#20](#)
- Notes: This is separated from empty search because the trigger is navigation/state reset, not text input validation.

BUG-FR05M-004: Very long search keyword breaks the mobile result-label layout

- Feature: FR-05-M Product Listing and Search Mobile
- Related test case: FR05M-BVA-006
- Severity: Medium
- Priority: Medium
- GitHub issue: [#21](#)
- Preconditions: Backend and React Native Mobile app are running.
- Steps to reproduce:
 1. Open the mobile app Home screen.
 2. Enter a 256-character search keyword.
 3. Tap Search.
 4. Observe the result label and product-list area.
- Expected result: The app remains responsive and displays the long keyword safely without breaking the layout.
- Actual result: The full long keyword is displayed in the result label, wraps into multiple lines, and breaks the layout by pushing/overlapping the product-list area.
- Evidence: [features/FR05_Product_Search_Mobile/evidence/BUG-FR05M-004.jpg](#)

- Notes: FR-05 does not define a maximum keyword length, so this is a robustness/UI layout defect rather than a strict boundary-limit defect.